

Software Requirements Specification (SRS) Template

Project: Hostel Management System

Version: 1.0

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Revision history

Version	Date	Author	Change summary	Approval
1.0	03-09-2026	Team 5	SRS for Hostel Management System	

Approvals

Role	Name	Signature / Email	Date
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Table of Contents

1. Introduction
2. Overall description
3. External interfaces
4. System features (detailed)
5. Non-functional requirements (detailed)
6. Quality attributes & Acceptance tests
7. UML Use-Case Diagram
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Hostel Management System. It defines functional and non-functional requirements, interfaces, and verification criteria intended for instructors and students to use as a reference.

1.2 Scope

Covers student hostel records, hostel and room management, room allocation, occupancy tracking, room transfer requests, complaint management, and administrative reporting. Excludes university-wide academic administration and payment processing beyond possible future APIs.

1.3 Audience

Developers, QA Engineers, Hostel Administrators, Wardens, System Integrators, and Assessment Evaluators.

1.4 Definitions

List of acronyms: HMS, UI, API, DB, RBAC, NFR, RTM, HTTPS, TLS.

2. Overall description

2.1 Product perspective

The Hostel Management System is a centralized application that manages student hostel details and room allocation. It includes a web-based user interface, application services, and a database for storing student, hostel, room, allocation, transfer, complaint, and administrative records.

2.2 Major product functions (detailed)

- User authentication and role-based access
- Student registration and profile management
- Hostel, block, and room management
- Room allocation
- Vacancy and occupancy tracking
- Check-in and check-out management
- Room transfer request management
- Complaint submission and tracking
- Administrative approval workflows
- Occupancy and vacancy reports
- Search and record retrieval
- Audit logging for important administrative actions

2.3 User roles and characteristics (expanded)

- Student: basic system usage knowledge; views hostel details and submits requests.
- Warden: monitors students, occupancy, complaints, and hostel activities.
- Hostel Administrator: manages student records, rooms, allocations, approvals, and reports.
- System Administrator: manages authorized user accounts and access roles.

2.4 Operating environment

Web application running on modern browsers, application server using Python

(Django/Flask) or PHP (Laravel), relational database such as MySQL, and a networked server environment.

2.5 Constraints

Unique student and room identifiers, room capacity rules, role-based access restrictions, secure handling of student information, and support for future integration through APIs.

3. External interface requirements

3.1 User interfaces

Primary UI: web-based dashboard with clear navigation for students, wardens, and administrators. Interfaces include login, student records, room allocation, occupancy, requests, complaints, and reports.

3.2 Hardware interfaces

- Desktop or mobile device with supported web browser
- Application server
- Database server
- Optional future integration with ID-card or biometric devices

3.3 Software interfaces

- Application framework: Python (Django/Flask) or PHP (Laravel)
- Relational database: MySQL
- Authentication and authorization components
- Optional notification or email service

3.4 Communications

- HTTPS/TLS for production client-server communication.
- Authenticated sessions or tokens for protected operations.
- Future external APIs shall use authenticated and documented request/response formats.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow HMS-F-####.

4.1 Authentication and Access Control

Description: Authenticate students and authorized staff and enforce role-based access to protected system functions.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-001	The system shall authenticate registered students and authorized staff using valid credentials.	Functional	High	Security / Hostel Admin	AC-HMS-F-001: Valid credentials create an authenticated session. Test: TC-Auth-01	Requires user account store
HMS-F-002	The system shall reject invalid login credentials and prevent access to protected functions.	Functional	High	Security	AC-HMS-F-002: Invalid credentials do not create a session. Test: TC-Auth-02	Authentication service required
HMS-F-003	The system shall enforce role-based access for Student, Warden, and Administrator functions.	Functional	High	Security / Hostel Admin	AC-HMS-F-003: Unauthorized role access is blocked. Test: TC-Auth-03	RBAC required

4.2 Student and Room Management

Description: Maintain student hostel records, hostel and room details, room capacity, occupancy, and availability.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-010	The system shall allow authorized staff to create and maintain student hostel records using a unique student ID.	Functional	High	Hostel Admin	AC-HMS-F-010: Valid student record is saved and duplicate ID is rejected. Test: TC-STU-01	Student ID uniqueness required
HMS-F-011	The system shall allow authorized staff to create and maintain hostel and room records including room capacity and availability.	Functional	High	Hostel Admin	AC-HMS-F-011: Room record is saved and displayed correctly. Test: TC-RM-01	Room capacity validation required

4.3 Room Allocation and Stay Management

Requirements: The system shall allocate available rooms to eligible students, prevent duplicate active allocations, record allocation dates, update occupancy, and support check-in/check-out. IDs: HMS-F-020..HMS-F-025.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-020	The system shall allow authorized staff to allocate an available room to an eligible student.	Functional	High	Hostel Admin	AC-HMS-F-020: Allocation links an eligible student to an available room. Test: TC-AL-01	Student and room must exist
HMS-F-021	The system shall record the room allocation date and allocation status for each student assignment.	Functional	High	Hostel Admin	AC-HMS-F-021: Allocation record contains date and status. Test: TC-AL-02	Allocation record required

HMS-F-022	The system shall prevent room allocation when the room occupancy has reached its configured capacity.	Functional	High	Business Rule	AC-HMS-F-022: Allocation is blocked when occupancy equals capacity. Test: TC-AL-03	Capacity validation required
HMS-F-023	The system shall prevent a student from having more than one active room allocation at the same time.	Functional	High	Business Rule	AC-HMS-F-023: Second active allocation is rejected. Test: TC-AL-04	Active allocation check required
HMS-F-024	The system shall allow authorized staff to record a student's hostel check-in.	Functional	Medium	Hostel Admin / Warden	AC-HMS-F-024: Check-in date and active stay status are recorded. Test: TC-AL-05	Depends on active allocation
HMS-F-025	The system shall allow authorized staff to record a student's hostel check-out and update room occupancy.	Functional	Medium	Hostel Admin / Warden	AC-HMS-F-025: Check-out updates stay status and room occupancy. Test: TC-AL-06	Depends on active stay

4.4 Room Transfer and Complaint Management

Requirements: Support student room transfer requests with approval or rejection workflows. Allow students to submit complaints and authorized staff to update complaint status and resolution details. IDs HMS-F-030..HMS-F-035.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-030	The system shall allow a student to submit a room transfer request with a reason.	Functional	Medium	Student	AC-HMS-F-030: Request is stored with pending status. Test: TC-TR-01	Student must be authenticated
HMS-F-031	The system shall allow authorized staff to approve or reject a room transfer request.	Functional	Medium	Warden / Admin	AC-HMS-F-031: Decision and request status are recorded. Test: TC-TR-02	Authorization required

HMS-F-032	The system shall update the student's room allocation and both room occupancy counts when an approved transfer is completed.	Functional	High	Hostel Admin	AC-HMS-F-032: Old and new room occupancy are updated consistently. Test: TC-TR-03	Transaction consistency required
HMS-F-033	The system shall allow students to submit hostel-related complaints.	Functional	Medium	Student	AC-HMS-F-033: Complaint is stored with a unique reference and status. Test: TC-CMP-01	Student authentication required
HMS-F-034	The system shall allow authorized staff to update complaint status and resolution remarks.	Functional	Medium	Warden / Admin	AC-HMS-F-034: Updated status and remarks are visible. Test: TC-CMP-02	Role restriction required

4.5 Reports, Monitoring and Audit

Requirements: Provide occupancy and vacancy reports, pending request views, complaint summaries, search facilities, and audit logging for important administrative actions. IDs HMS-F-040..HMS-F-045.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-040	The system shall generate occupancy and vacancy summaries for hostel rooms.	Functional	High	Hostel Admin / Warden	AC-HMS-F-040: Report totals match current allocation data. Test: TC-RPT-01	Accurate room data required
HMS-F-041	The system shall allow authorized staff to view pending room transfer requests and unresolved complaints.	Functional	Medium	Hostel Admin / Warden	AC-HMS-F-041: Pending items are correctly listed by status. Test: TC-RPT-02	Status filtering required

HMS -F- 042	The system shall maintain an audit record for important administrative actions such as allocation changes and room updates.	Functional	High	Security / Admin	AC-HMS-F-042: Audit record contains user, action, and timestamp. Test: TC-AUD-01	Audit logging required
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5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs HMS-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
HMS-NF-001	Common user operations shall complete within 3 seconds for 90% of requests under normal load.	Performance	High	90th percentile $\leq$ 3s. Test: TC-Perf-01
HMS-NF-002	System shall provide 99% monthly availability excluding scheduled maintenance.	Reliability	High	Uptime reports show $\geq$ 99% per month. Test: TC-REL-01
HMS-NF-003	Student and room allocation data shall remain consistent and room capacity shall not be exceeded.	Data Integrity	High	Integrity tests pass. Test: TC-DATA-01
HMS-NF-004	System shall provide responsive and understandable interfaces on supported desktop and	Usability	Medium	Usability test pass. Test: TC-UX-01

	mobile browsers.			
HMS-NF-005	System data shall be backed up according to the configured backup schedule and restoration shall be possible.	Recovery / Maintainability	High	Backup restoration verified. Test: TC-REC-01

5.1. Security

5.1.1 Security Objectives

Security objectives for the Hostel Management System are to protect student and hostel data from unauthorized access and to ensure that important administrative actions are authorized and auditable.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
HMS-SR-001	The system shall require authentication before granting access to protected student or administrative functions.	Security	High	Unauthenticated access is denied. Test: TC-SEC-01
HMS-SR-002	The system shall enforce role-based authorization for Student, Warden, and Administrator functions.	Security	High	Unauthorized role access is blocked. Test: TC-SEC-02
HMS-SR-003	The system shall store passwords using secure one-way hashing and shall not store plaintext passwords.	Security	High	No plaintext passwords found. Test: TC-SEC-03
HMS-SR-004	The system shall protect production client-server communication using HTTPS/TLS.	Security	High	Protected communication uses HTTPS. Test: TC-SEC-04

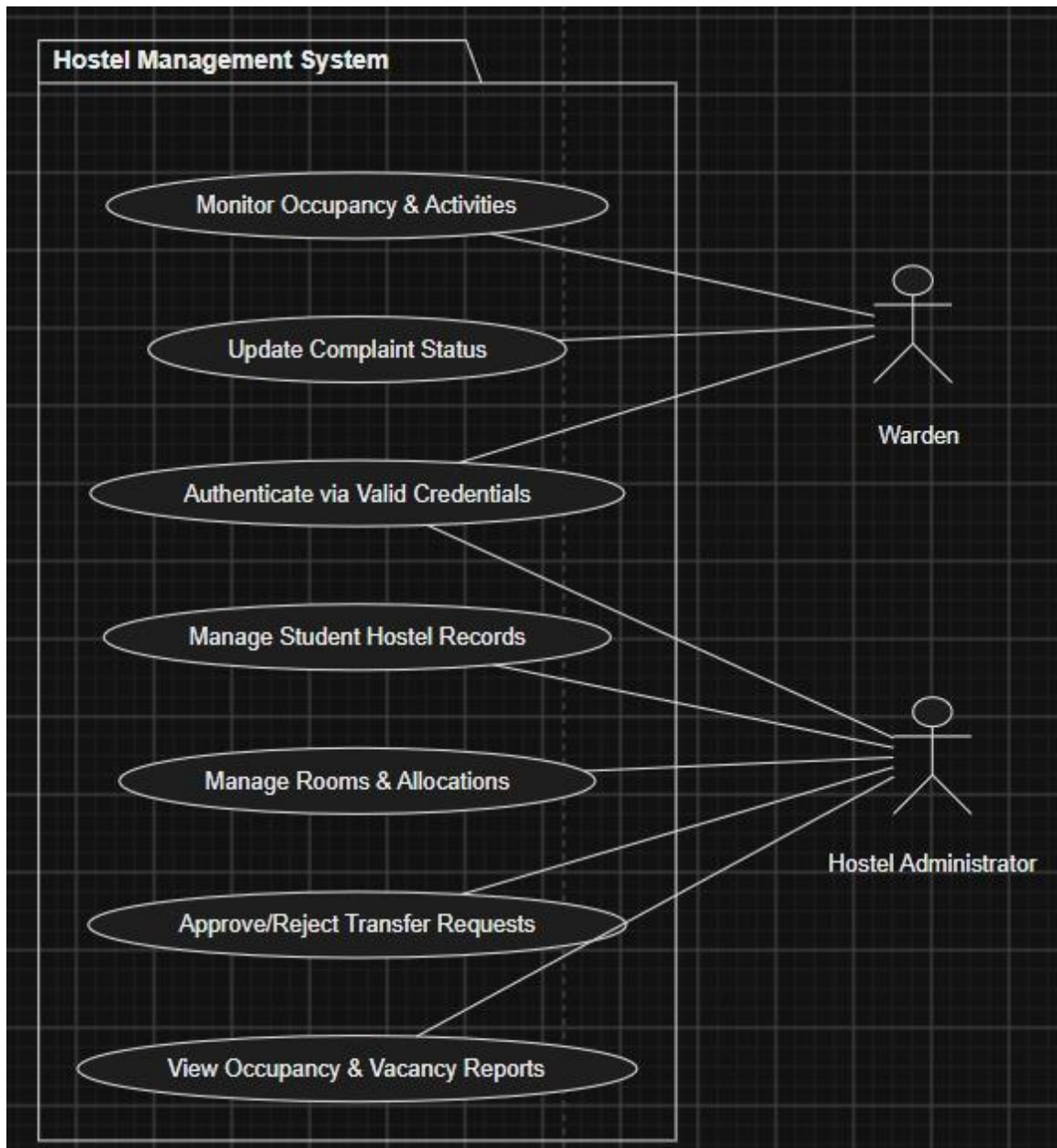
HMS-SR-005	The system shall validate and sanitize user inputs to reduce injection and malformed-input risks.	Security	High	Security input tests pass. Test: TC-SEC-05
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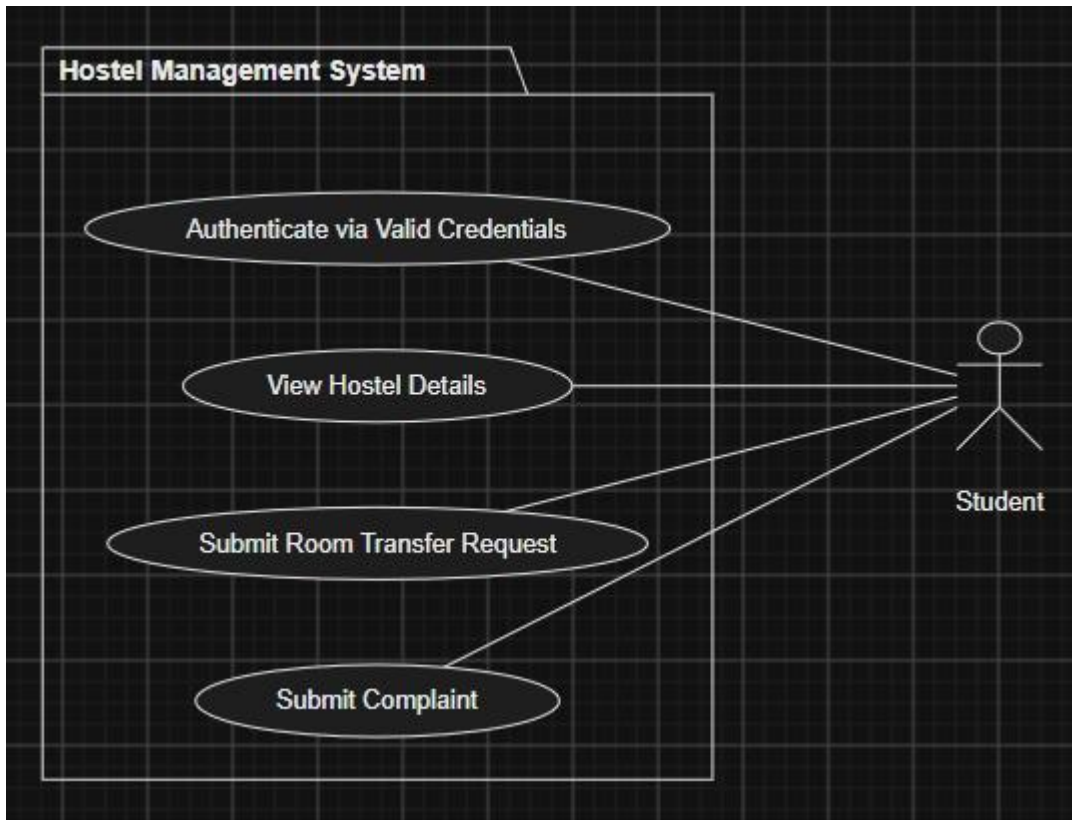
6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR or security failures, and RTM shows all test cases passed.
- Acceptance test suites: Authentication, Student Management, Room Management, Allocation, Transfer, Complaint, Performance, Security, and Usability tests.

7. System models and diagrams

7.1 UML Use-Case diagram





8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
HMS-F-001	User login	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
HMS-F-010	Student record	4.2 / DS-Student-01	Student Module	TC-STU-01	N	
HMS-F-020	Allocate room	4.3 / DS-Allocate-01	AllocationModule	TC-AL-01	N	